

Electrical Machinery

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SID 810103540

1-

$$v_{rms} = 4.44 \times B \times f \times N \times A$$

$$230 = 4.44 \times 0.9 \times 50 \times N \times A$$

$$1.51 = N \times A$$

if we consider A as square area $A = 36 \times 10^{-4} m^2$

so N is equal to 320 for the HV side and 55.6 or 56 for the LV side.

$$l = 2f + 2b - e$$

$$l = 0.54m$$

- 2- For calculating Reluctance in Simulink , first we determine magnetic flux , then calculate magnetic field , then calculate magnetic field strength at last we can determine mmf.

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3- function mmf = variable_rel(mflux)
4-     % Parameters
5-     A = 36 * 10^(-4);    % Area in m^2
6-     Length = 0.54;      % Length in meters
7-
8-     % 1. Calculate Flux Density (B = Phi / A)
9-     B = stomach_muscles(mflux) / A;
10-
11-     H = 0.1431 * exp(6.253 * B) - 0.1431;
12-     mmf = H * Length * sign(mflux);
13- end

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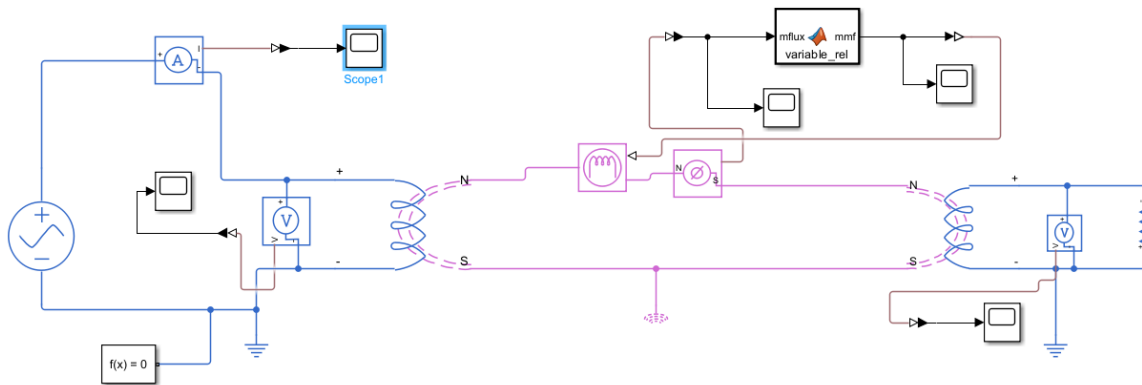


Figure 1_circuit schematic

As you can see unlike current flux is sinusoidal the reason is behind hysteresis loss and B_H curve

$$Ni = H.l \rightarrow i = \frac{H.l}{N}$$

So, current also act like H .

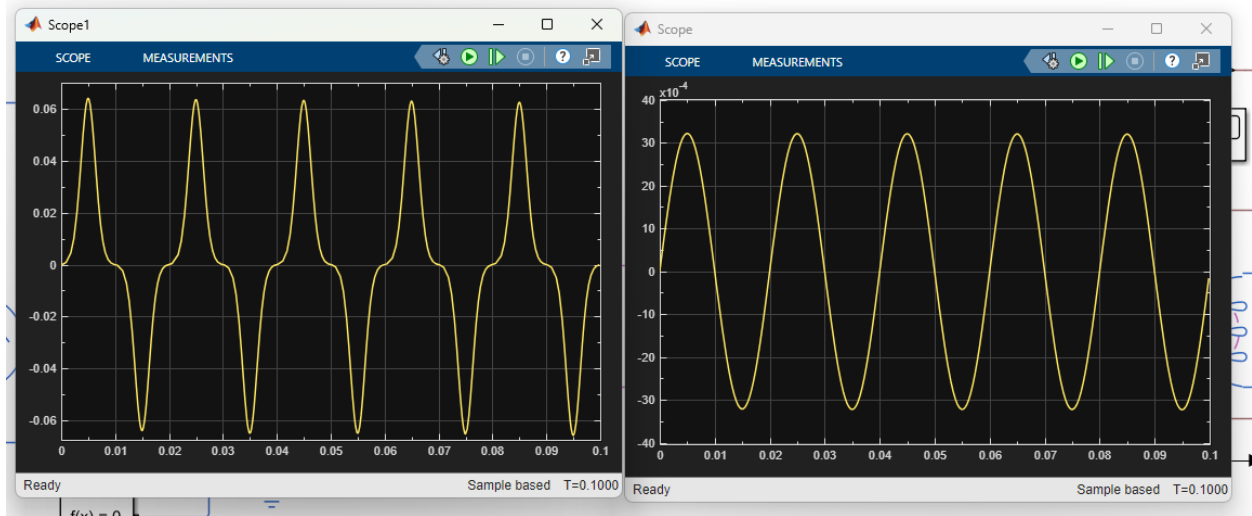


Figure 2_current in scope 1 and flux in scope

3 –

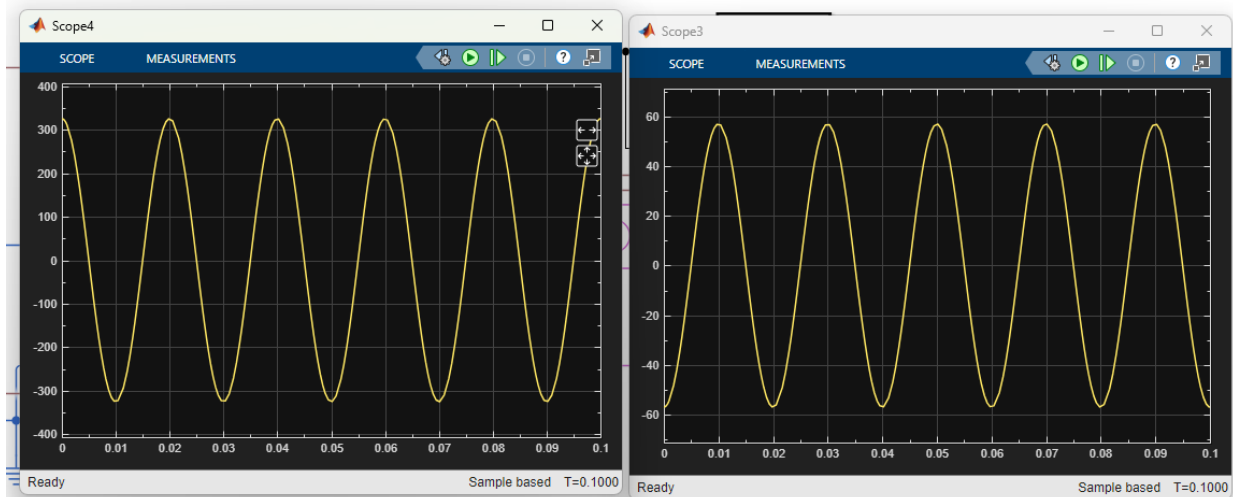


Figure 3_ voltages over transformers

Scope 4 is corresponding to HV and Vpeak is 324.8 , for the LV it's 56.88

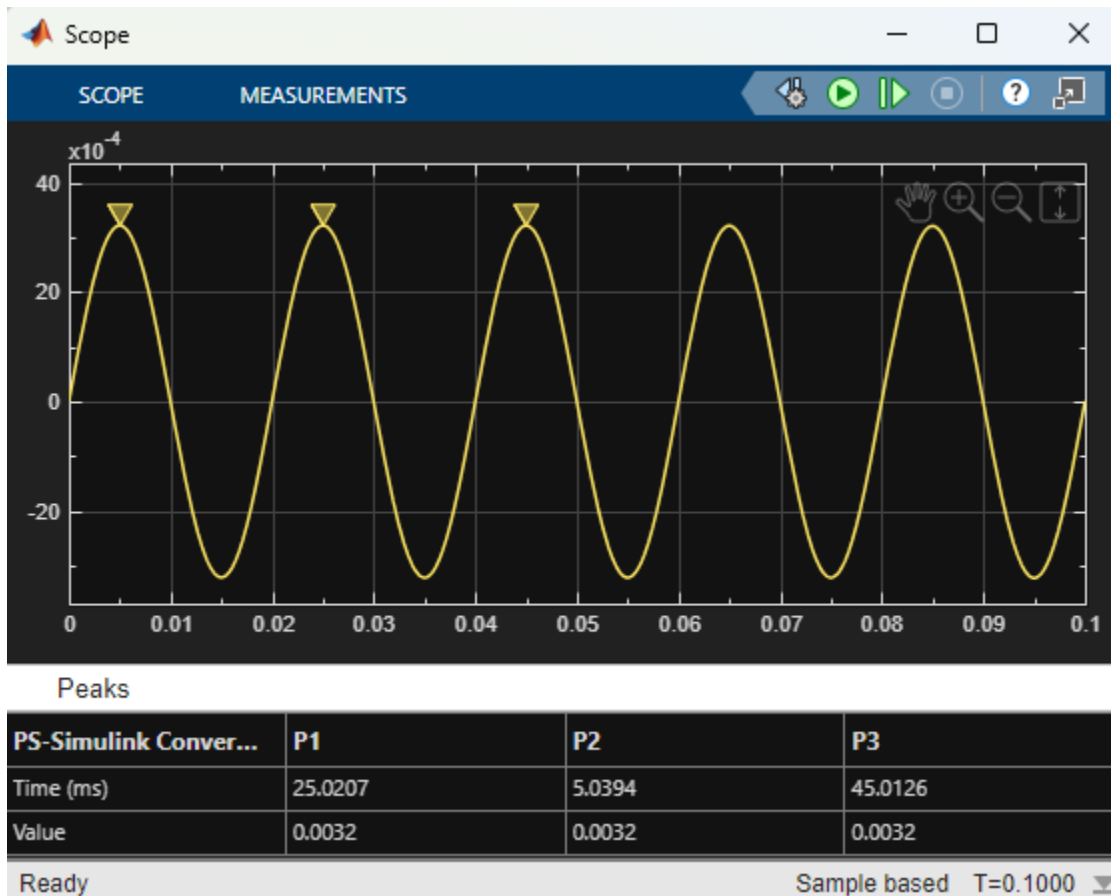
Which correspond to $\frac{n1}{n2} = \frac{230}{40}$

4 –

$B = \frac{\phi}{A}$ by calculating flux peak in figure 2 which is 0.0032 , A = 0.0036, so B = 0.89 which is acceptable .

5- by assuming $R1 = 8 \text{ ohm}$ and $R2 = 0.2 \text{ ohm}$ we can see that flux act sinusoidal but actually it's not because 'I' changes non_sinusoidally

$$\phi = \int v \cdot dt \rightarrow v = v_{in} - Ri$$



6-



Figure 4_ currents



Figure 5_ voltages

$$0.9537 * \frac{230}{40} = 5.483 \quad v_{rms1} = 229.9, v_{rms2} = 38.07$$

$$325.239 * \frac{40}{230} = 56.55 \quad I_{rms1} = 0.674, I_{rms2} = 3.854$$

So, with a low approxiamtion it's good to say that our transformer ratio is correct but if I want to be precise loss on resistors in series with load and voltage source cost change in transformer voltage.

All of the figures are sinusoidal . the reason that current is sinusoidal too is because of when we have load we have a load current which is massively bigger than hysteresis current and it's a function of V_{in} so it's sinusoidal.

7-

$$R = \frac{l}{\mu A} = \frac{0.1 * 10^7}{4\pi * 0.0625} = 1,273,239$$

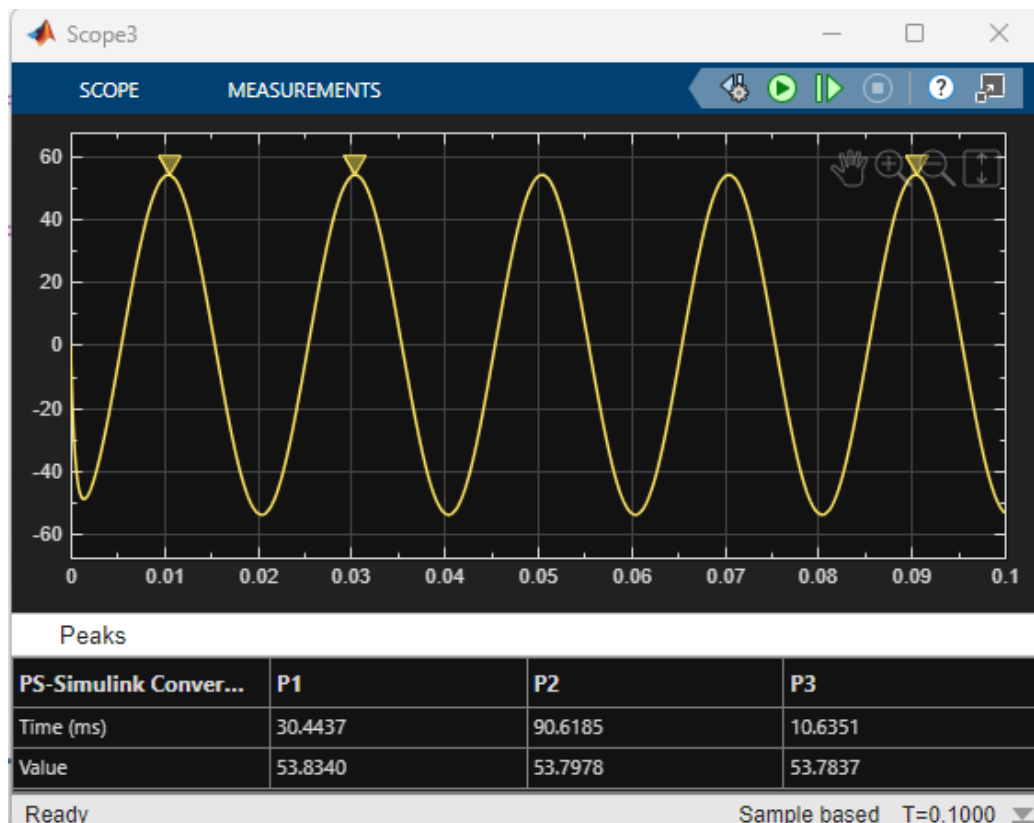


Figure 6_transformer Voltage2

Yes, V_2 changed because now flux divides , some of it goes through the reluctance and by the faraday's law V_2 reduces as the flux does.

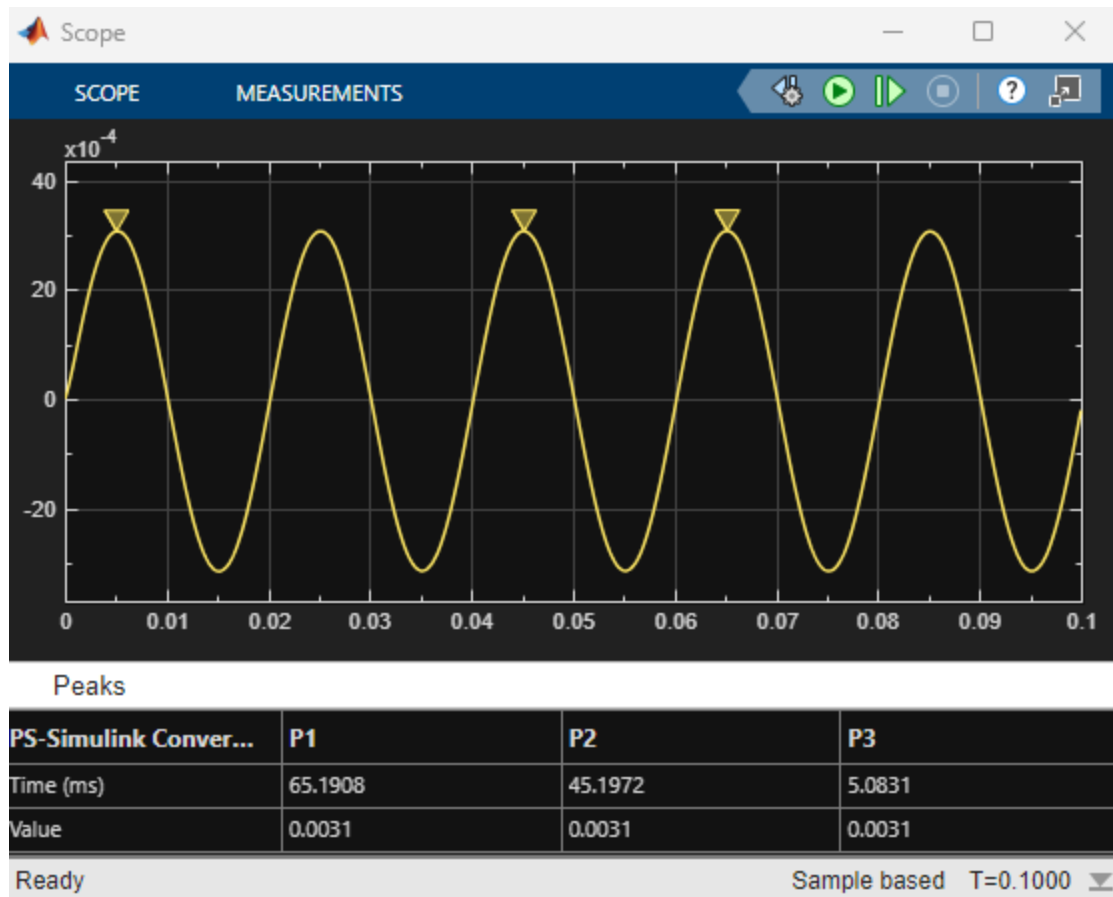


Figure 7_flux

Flux flow also reduced , when input voltage flux divides between the main path and the reluctance , so it will be reduced.

8- $R_{load} = 100k$

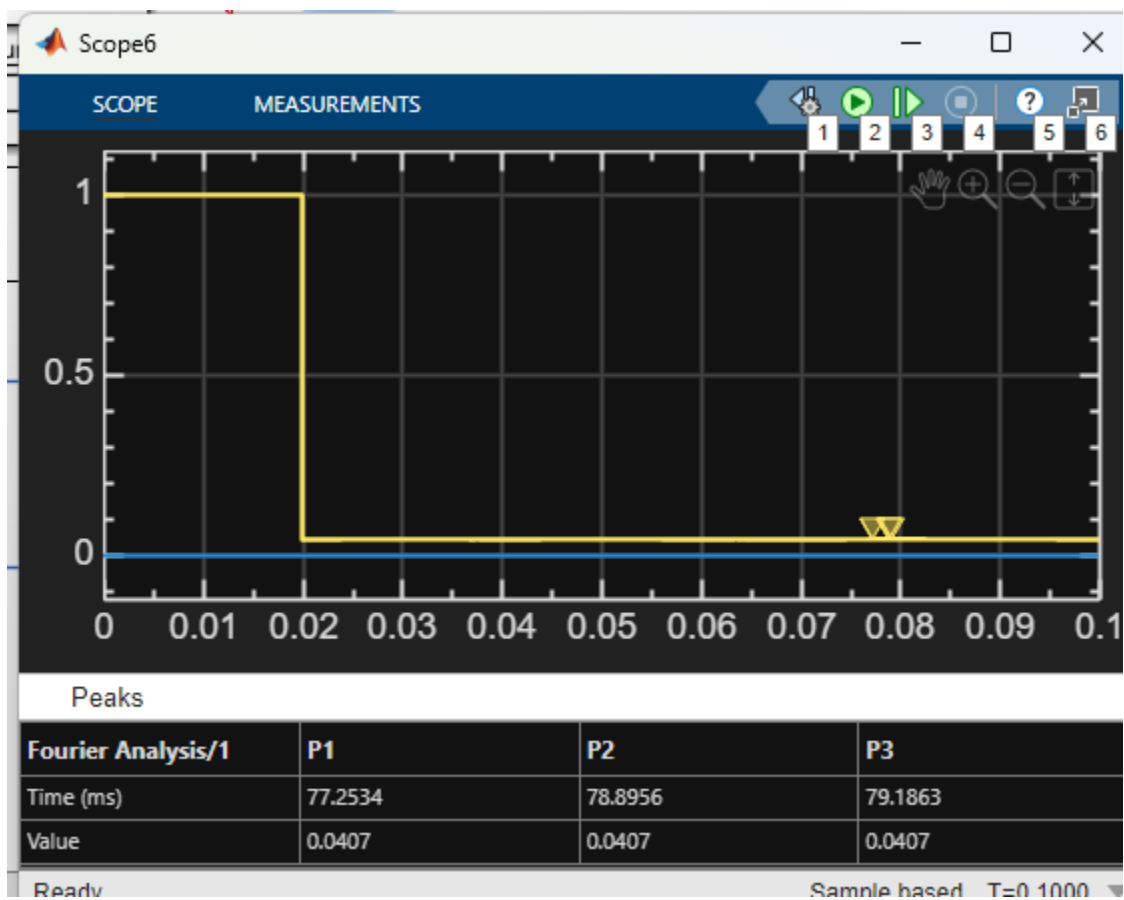
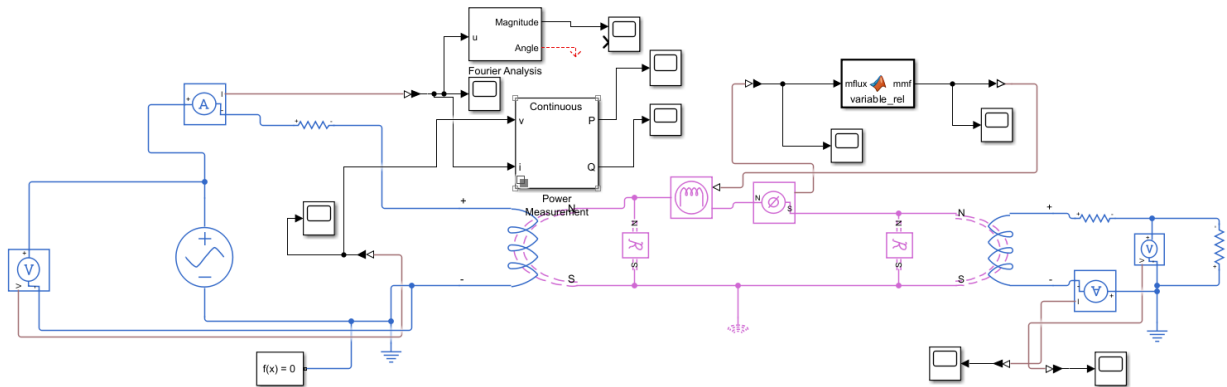
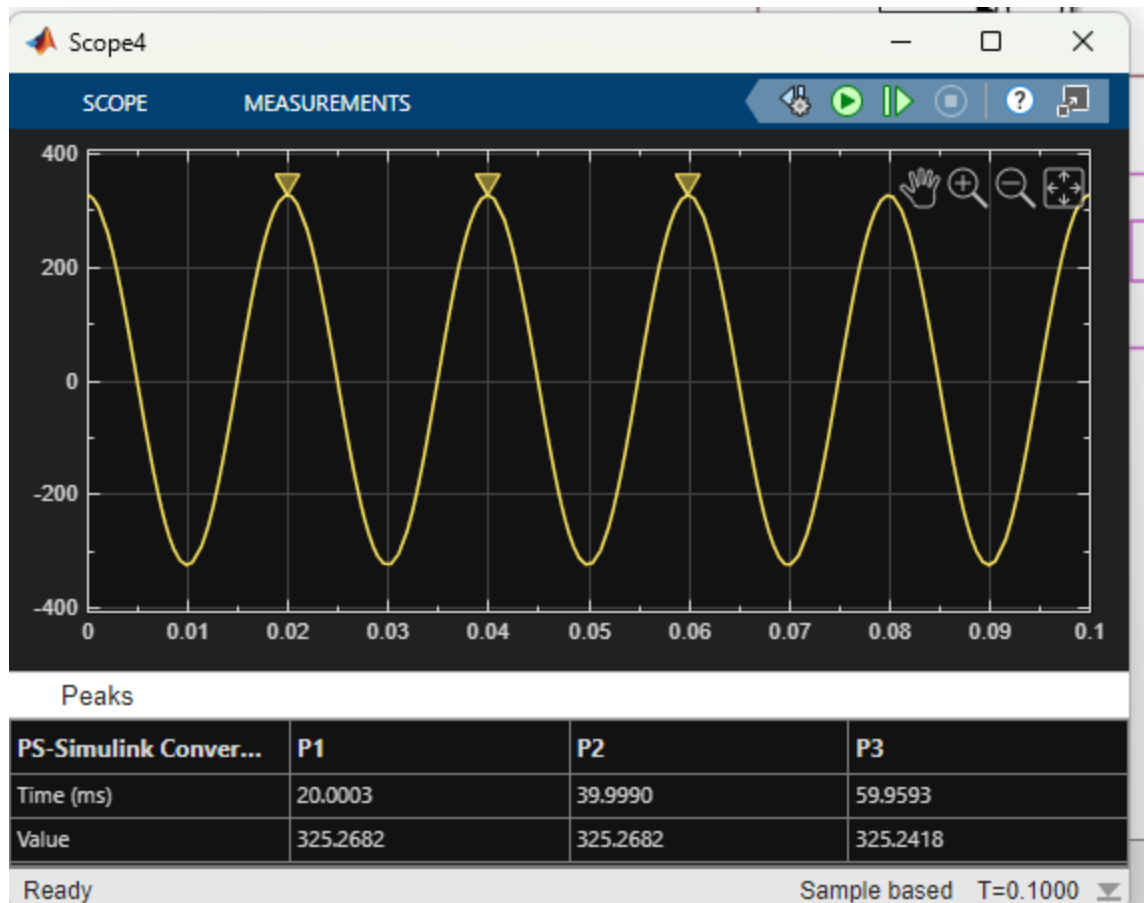


Figure 8_current in open circuit



$$V_{\text{rms}} = 229.8.$$

$$I_{\text{rms}} = 28.779 \text{ mA}$$

For calculations of current we get the current goes through the fourier block by setting $f = 50\text{Hz}$ and $n = 1$. which I_{rms} is equal to 28.779 mA.

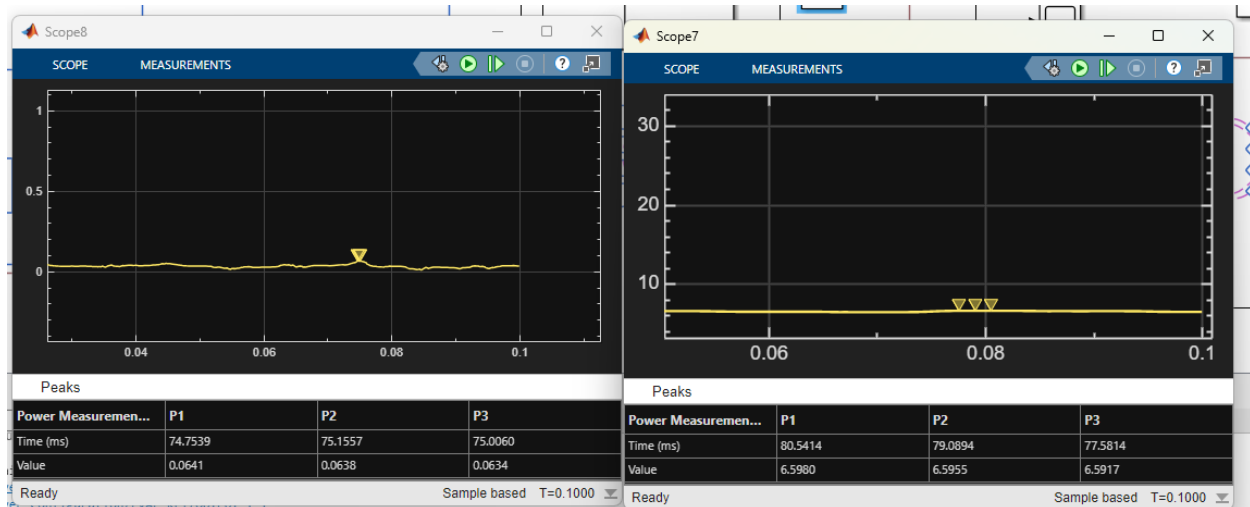


Figure 9_ power in open circuit scope 8 is p & scope 9 is Q

Open Circuit Test:

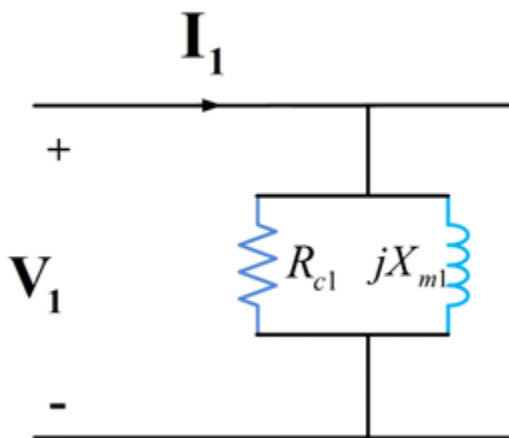


Figure 10_ equivalent circuit for OC

In power block I calculate only the first harmonic for 50Hz

According to figure 10 active power is (scope 8) 0.064 watt and reactive power(scope 7) is 6.59

$$R_{OC} = \frac{V^2}{P} = 826,562\Omega$$

$$X_{OC} = \frac{V^2}{Q} = 8,027 \rightarrow Z = j\omega L \rightarrow L = 25.55H$$

Does $N^2 / RL = XOC = Lm$?

$N = 320$, mmf Domain = 12.887(mmf wasn't sinusoidal so I considered the first harmonic), flux domain = 0.0032 , Reluctance in work zone = 4027.

$N^2 / RL = 25.42$ which is almost close to the number we measured theoretical.

Short circuit test:

Nominal current is : 0.942 A = 0.667(rms) , and assuming $R_{load} = 0.00001\Omega$.

Change input voltage to reach the nominal current it happens for $V_{in} = 49.9 = 35.3v_{rms}$



Figure 11_power in SC Test

According to figure 10 active power is (scope 8) 6.55 watt and reactive power(scope 7) is 22.71

$$R_{SC} = \frac{P}{I^2} = 14.72\Omega$$

$$X_{SC} = \frac{Q}{I^2} = 51 \rightarrow Z = j\omega L \rightarrow L = 0.162H$$

Does $N^2 / R_L + N^2 / R_{L2} = X_{SC} = X_{e1,2}$?

$N = 320, 56$.

$R_L = 1,273,239(Q7)$

$X_{SC} = 0.0828$

Which isn't equal to the number we measured .